BrainWave5

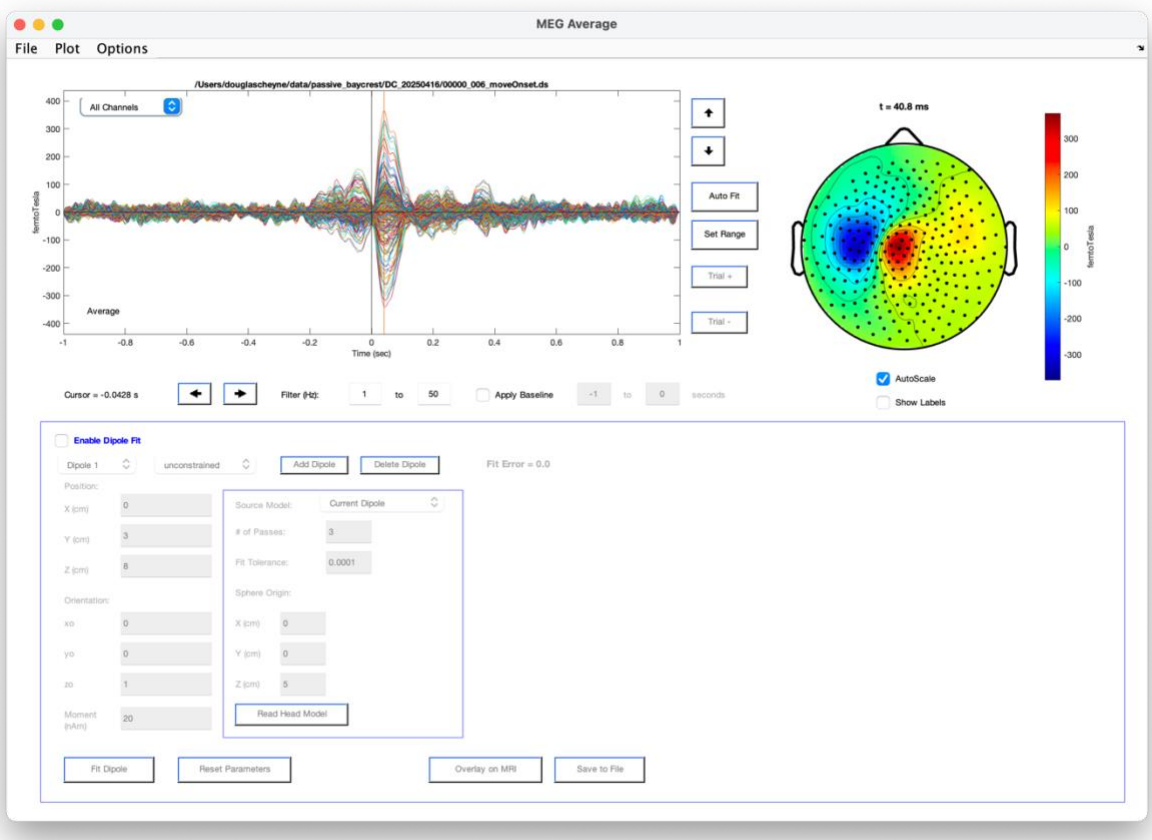
Data Plot / Dipole Fit User Guide

The DataPlot module in BrainWave5 provides a GUI driven plotting tool for viewing a time – amplitude plot of single trial or multi-trial data. After creating an epoched dataset it will be automatically be displayed in the DataPlot / Dipole Fit window. Alternatively, you can open any epoched (or single trial) dataset in Data Plot from the Main Menu.

By default the file will be opened showing the average as an overlay of all channels ("butterfly plot") with a latency cursor place at the time point of maximum amplitude and the field pattern at the sensors (topoplot) at that time point.

The lower Dipole Fit panel can be used to fit single or multiple dipole models to selected time points in the data.

The example below shows a butterfly plot of an averaged MEG response time-locked to a right finger movement showing a dipolar field pattern over the left sensorimotor cortex.



Menu Options

File	
Load CTF Dataset…	Loads an epoched or single trial MEG dataset.
Close	Closes the current window.

Plot	
Plot Average	Plots the averaged sensor data for all channels.
Overlay All Trials	Overlays all trials sensor when displaying a single channel. To enable select one channel display dropdown.
Plot Single Trial	Plots only one trial sensor data for a single channel. Enabled if one channel selected from display dropdown.
Plot Global Field Power	Plots the RMS amplitude averaged over all sensors as a single, rectified waveform.

Options	
Edit Channel List…	Opens a GUI to visually select or remove channels from the plot.
Plot Single Channel TFR	Create a time-frequency representation (TFR) plot. Enabled when viewing a single channel selected from display dropdown.

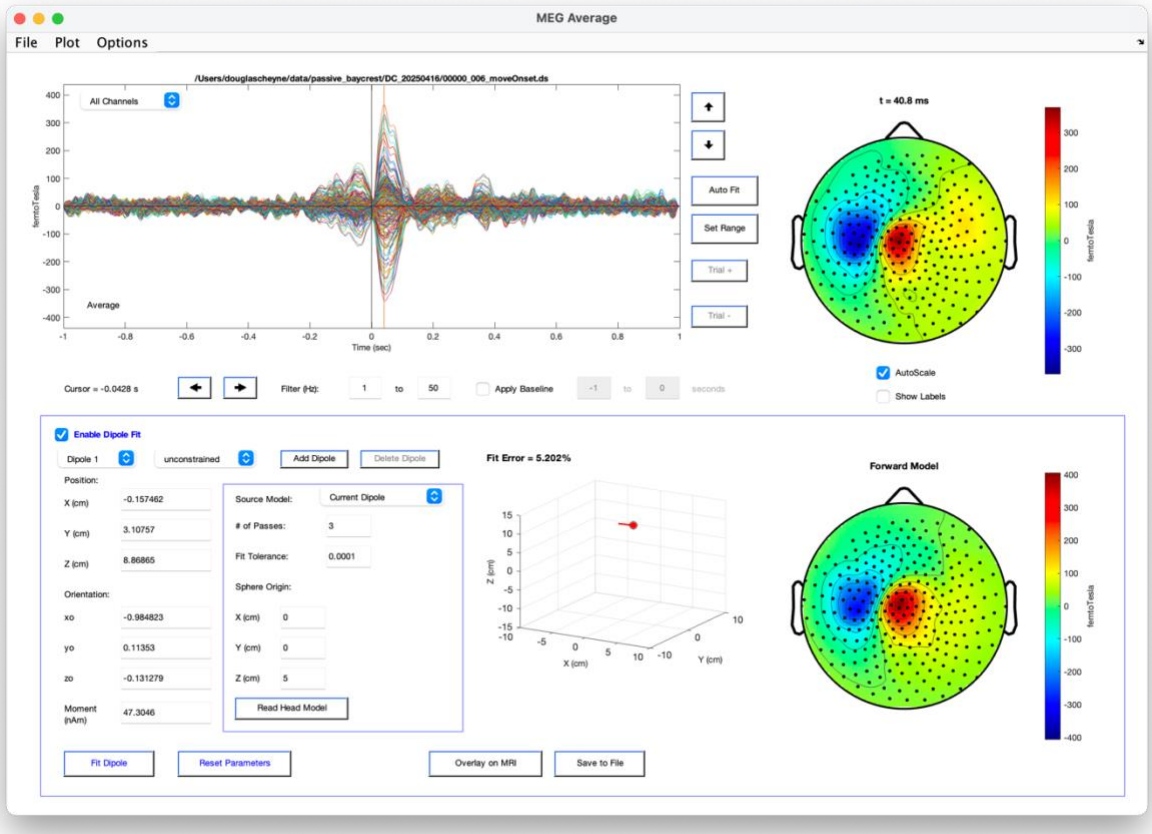
Display Options

Arrow Keys: left/right arrows move latency cursors, up/down arrows to increase/decrease amplitude scale.
Auto Fit: autoscales the currently displayed data.
Set Range: edit text fields to set display amplitude range.
Filter: edit text boxes to set low and high pass filter settings.
Baseline: enable and edit text fields to apply / adjust the baseline (DC offset).
Show Labels: enable to display channel names on the topoplot.

Dipole Fit

The DipoleFit panel below the channel plot can be enabled (check box) and used to perform least-squares fitting of the location, orientation and amplitude of single or multiple equivalent current dipoles to a the time point currently selected in the DataPlot window. This uses a Simplex minimization algorithm to fit one or more dipoles to the selected time point with options to constrain dipole location(s) or fit mirror symmetric dipoles. After selecting start parameters and time point click on Fit Dipole. The fit error is displayed next to the 3D plot of the dipole location along with the forward modeled field pattern.

Example below shows single current dipole fit to the latency of the movement evoked field at 40.8 ms showing 5.2% fit error.



Start parameters: Position and orientation can be manually entered for each dipole in the labelled text fields. Unconstrained = fit all dipole parameters. Constrained = fix dipole location to start position (fit orientation and amplitude only). Mirror Symmetic = constrain dipoles to be always bilaterally symmetric.

Source Model: menu to select the type of dipole source. Current dipoles for brain sources or magnetic dipoles (e.g., magnetic dipole phantom data).
Read Head Model: Loads single sphere or multisphere models from a Head Model (*.hdm) file.
Add / Delete Dipole: to add or removed additional dipoles to the model.
Overlay on MRI: Click to automatically open **MRI Viewer** with an overlay of the fitted dipole(s) on the subject's MRI.
Save To File: Saves fitted dipole parameters saved to a tagged text file (*.dip) (compatible with the CTF .dip file format) which can also be opened directly in **MRI Viewer** from the Overlay menu.
Fit Tolerance / # of passes: Adjusts the error tolerance and number of re-initializations of the Simplex algorithm for optional fitting (default values recommended).